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June 24, 1966
SUPERSEDING
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October 29, 1964

INTERIM FEDERAL SPECIFICATION

COMPRESSOR, PISTON RING, BAND-TYPE

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Standardization Division, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized Federal agencies to use this Interim Federal Specification as a valid exception to Federal Specification GGG-C-555 dated December 8, 1955.

1. SCOPE AND CLASSIFICATION

1.1 Scope. The piston ring compressors covered by this specification are used to firmly compress the piston rings into the grooves of the piston to facilitate insertion of the piston and rings into a cylinder.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity indicated by the titles of the specifications, but cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Sizes. Piston ring compressors shall be of the following sizes, as specified (see 6.1):

Size 0. - 1 1/2 to 3-inch capacity.
Size 1. - 2-1/8- to 5-inch capacity.
Size 2. - 3-1/2- to 7-inch capacity.
Size 2L.- 3-1/2- to 7-inch capacity. (longbody see 3.3.1 and 3.9).
Size 3. - 6 to 9-inch capacity.
Size 4. - 9 to 12-inch capacity.
Size 5. -12 to 15-inch capacity.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specifications:

MIL-H-15424 - Hand Tools; Packaging of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustration. The illustration shown herein is for convenience of identification and does not preclude purchase of piston ring compressors otherwise conforming to this specification.

3.2 Material. The material used in construction of the piston ring compressors shall be as hereinafter specified. The material shall be free from any defects and imperfections that may affect their serviceability. When materials are used which are not specifically designated, they shall be entirely suitable for the purpose intended.

3.3. Design. The size 0 piston ring compressors shall consist essentially of a compression band, tension band, spool, removable key or wrench, and either a ratchet and pawl or other anti-reverse means at the option of the manufacturer, and a friction brake; sizes 1, 2, 3, 4, and 5 shall have the same components except two tension bands shall be furnished; and size 2L shall have 4 tension bands. The same piston ring compressors shall be designed to accomplish the following:

(a) To compress the heaviest piston rings serviced by means of band tension providing equalized pressure over the width of the compression band.

(b) To provide the compression force by means of a spool, actuated by either a removable key or wrench.

(c) To lock in position at any desired point within the operating range by means of either a ratchet and pawl, or other anti-reverse means (see 3.3.5 and 3.3.6).

(d) To provide takeup tension of approximately 0.010 inch per tooth on ratchet, or to lock in position automatically at any point within the operating range of the compressor.

(e) To provide a friction brake for controlling opening feature.

(f) To provide suitable means on the lower edge of the compression band to prevent the compression band from entering cylinder when piston is inserted.

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3.3.1 Compression band. The compression band shall be of good quality spring steel, properly tempered so as to fully open the completely assembled compressor to the largest required diameter when not restrained. The width of the compression band shall be not less than 3-1/2 inches, for sizes 1, 2, 3, 4 and 5, not less than 2 inches for size 0, and not less than 6 1/2 inches for size 2L.

3.3.2 Tension bands. The tension band(s) shall be of good quality spring steel, properly tempered, and of ample strength to compress the heaviest rings to be serviced. Means shall be provided for proper assembly of the tension band(s) to the spool.

3.3.3 Spool. The spool shall be of good quality steel, designed to properly accept and wind the tension bands without slippage. Means shall be provided in the spool to properly accept a removable key or wrench which actuates the spool.

3.3.4 Removable key or wrench. The key or wrench provided to actuate the spool shall be of good quality steel and of sufficient length and strength to actuate the spool in contracting the bands and piston rings to the diameter desired.

3.3.5 Ratchet and pawl design. When a ratchet and pawl design is furnished, the ratchet shall be of good quality steel, properly heat-treated to resist wear or breakage under normal service conditions. The ratchet shall be designed for proper assembley to the spool and shall have the teeth so spaced as to accomplish a takeup of approximately 0.010 inch per tooth. The pawl shall be of good quality steel, properly heat-treated to resist wear or breakage under normal service conditions. The pawl shall be designed to be properly assembled to the tension bands and to lock the ratchet at the desired point.

3.3.6 Other anti-reverse means. When an anti-reverse means (other than ratchet and pawl) is furnished, a spring clutch or other mechanism which will prevent reverse rotation of the spool except under a released condition will be acceptable. A steel button or pin-type handle shall be provided for releasing the anti-reverse spring clutch or mechanism.

3.3.7 Friction brake. A friction brake shall be provided to prevent the compression band from immediately springing open upon release of the ratchet and pawl or other anti-reverse means. The brake shall consist of a spring steel clamp or adjustable friction clamp arranged around the spool so as to erase rotation.

3.4 Finish. The piston ring compressors shall have a smooth finish, free from burrs, roughness, and excessive oxide scale.

3.5/ Identification marking. The piston ring compressors shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.6 Size 0, 1 1/2 to 3-inch capacity. Size 0 piston ring compressors shall be capable of servicing pistons ranging from 1 1/2 up to and including 3 inches in diameter and shall be similar to figure 1.

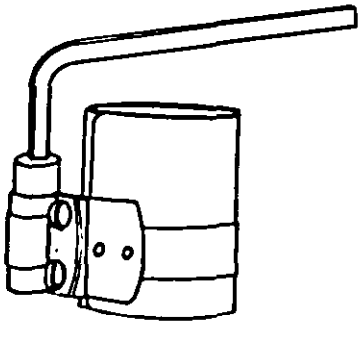


Figure 1. - Piston ring compressor. (Size 0)

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3.7 Size 1, 2-1/8- to 5-inch capacity. Size 1 piston ring compressors shall be capable of servicing pistons ranging from 2-1/8 up to and including 5 inches in diameter, and be similar to figure 2.

3.8 Size 2, 3-1/2- to 7-inch capacity. Size 2 piston ring compressors shall be capable of servicing pistons ranging from 3-1/2 up to and including 7 inches in diameter, and shall be similar to figure 2.

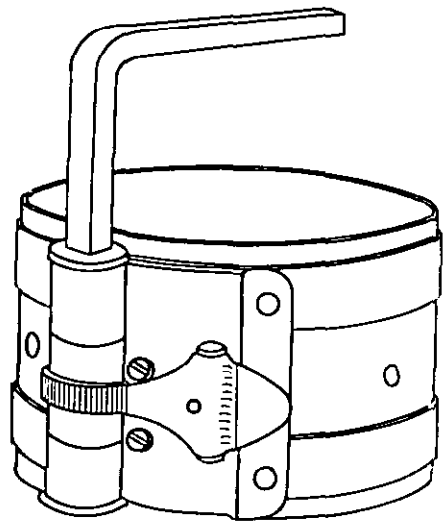


Figure 2. - Piston ring compressor. (Sizes 1, 2, 3, 4, and 5)

Note: Ratchet and pawl design shown, other designs are optional (see 3.1 and 3.3.6)

3.9 Size 2L, 3-1/2 to 7-inch capacity (long body). Size 2L piston ring compressors shall be capable of servicing piston rings ranging from 3-1/2 up to and including 7 inches in diameter, and shall be similar to figure 3.

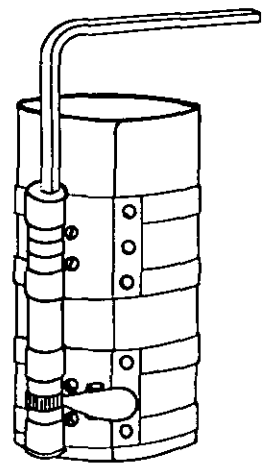


Figure 3. - Piston ring compressor (Size 2L)

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3.10 Size 3, 6 to 9-inch capacity. Size 3 piston ring compressors shall be capable of servicing piston rings ranging from 6 up to and including 9-inches in diameter, and shall be similar to figure 2.

3.11 Size 4, 9 to 12-inch capacity. Size 4 piston ring compressors shall be capable of servicing piston rings ranging from 9 up to and including 12-inches in diameter and shall be similar to figure 2.

3.12 Size 5, 12 to 15-inch capacity. Size 5 piston ring compressors shall be capable of servicing piston rings ranging from 12 up to and including 15-inches in diameter and shall be similar to figure 2.

3.13 Workmanship. Workmanship shall be of the highest grade throughout and equal in every respect to the best commercial practice. The piston ring compressors shall be free from rust, burrs, fins, and any imperfections which may impair their serviceability or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4. 2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.2.1 Inspection of materials and components. The supplier is responsible for insuring that the materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2.2 Inspection of the end item. Inspection of the compressors shall be performed as specified in the following paragraphs. A lot shall consist of compressors of the same size. The sample unit shall be one completely fabricated compressor.

4.2.2.1 Visual and dimensional examination. Each sample unit shall be visually examined for the defects set forth in table I. The inspection level shall be level II, with an acceptable quality level of 2.5 defects, expressed in terms of defects per hundred units.

Table I.- Classification of defects

Examine	Defect
Finish	Indication of rust.
	Not smooth, burrs, oxide scale.
Design	Characteristics not in accordance with specified requirements.
Construction and Workmanship	Operation omitted or not properly performed.
	Tension band(s) not securely joined to spool.
	When applicable, ratchet mechanism not securely joined to spool.
	When applicable, teeth of ratchet, missing, broken or damaged.
	Rivets or screws missing or loose.
	Key or wrench missing, bind,s not easily inserted or removed from spool.

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	Lower edge of compression band not provided with means to prevent band from entering cylinder.
	When applicable, the anti-reverse spring clutch or its release button is missing or ineffective.
	Friction brake missing or ineffective.
Marking	Missing, incorrect, or illegible.

4.3 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing and marking comply with the requirements in Section 5 of this specification. Defects shall be scored in accordance with table II. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

Table II Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, or distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Unless otherwise specified (see 6.1), the piston ring compressor shall be preserved, packaged and packed in accordance with MIL-H-15424. The level of preservation and packaging shall be A or C and the level of packing A, B or C as specified (See 6.1).

5.1.1 As part of the requirements of Section 5 of this specification the following is considered as part of Table I of MIL-H-15424.

Column 1	Column 2	Column 3	Column 4	Column 5	Column 6
Compressor, piston ring, Band type	P-8, P-9	1	1	6	10

5.2 Marking.

5.2.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military activities. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data. Purchasers should select the preferred options offered herein and include the following data in procurement documents:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2.1).
(c) Level of preservation, packaging and packing required (See 5.1).
(d) Marking, if different (see 5.2).

6.2 For civil agency procurements. When level B packaging is required for civil agency procurements, the level A requirements in 5.1 shall apply.

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